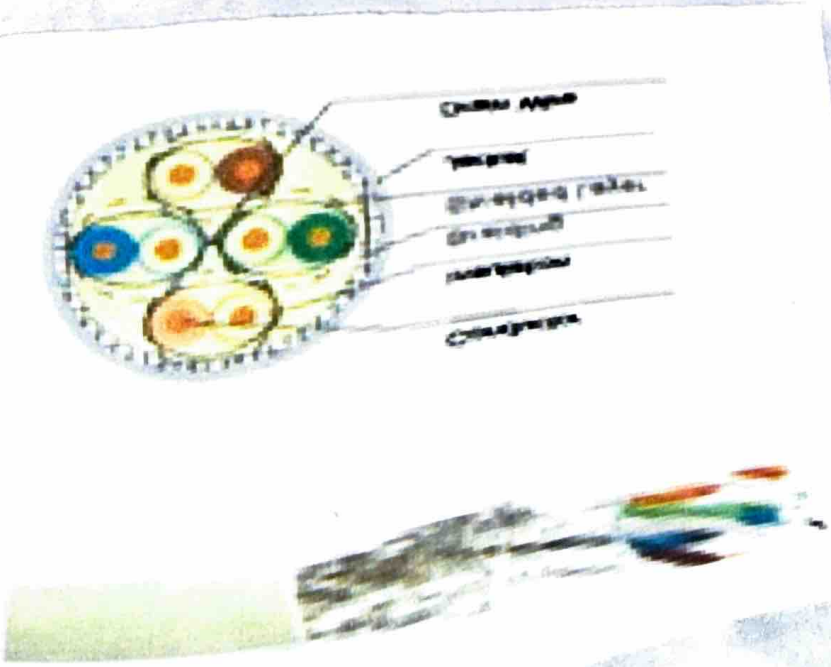


Twisted pairs with color-coded plastic insulation

Outer jacket



16/7 Exp 2.

STUDY OF DIFFERENT TYPES OF NETWORK CABLES

Aim: To study and understand different types of network cable, to make your own ethernet cross over cable / straight cable.

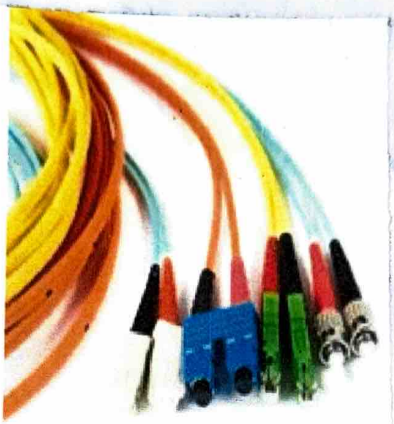
Different Types of Network cable:

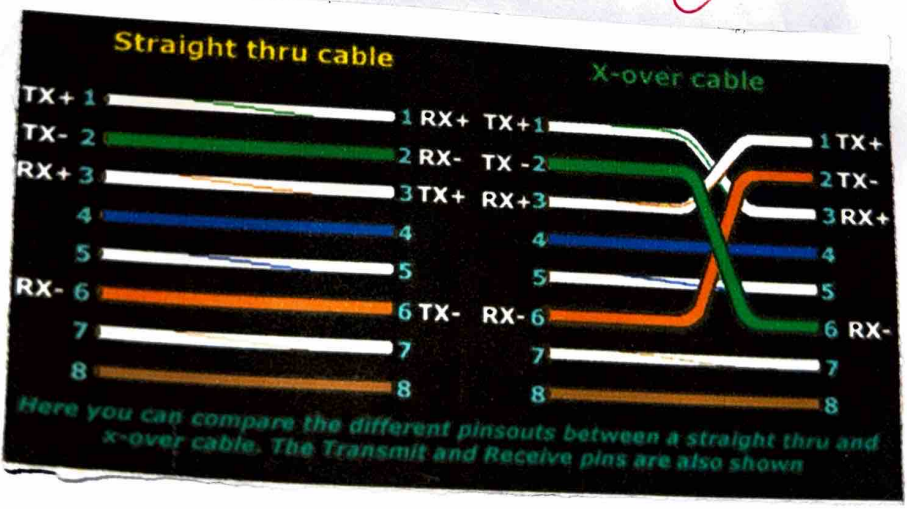
1) unshielded Twisted pair cable
2) Shielded Twisted pair cable
3) Coaxial cable
4) Fibre optic cable

UTP: In unshield Twisted pair cable has three categories

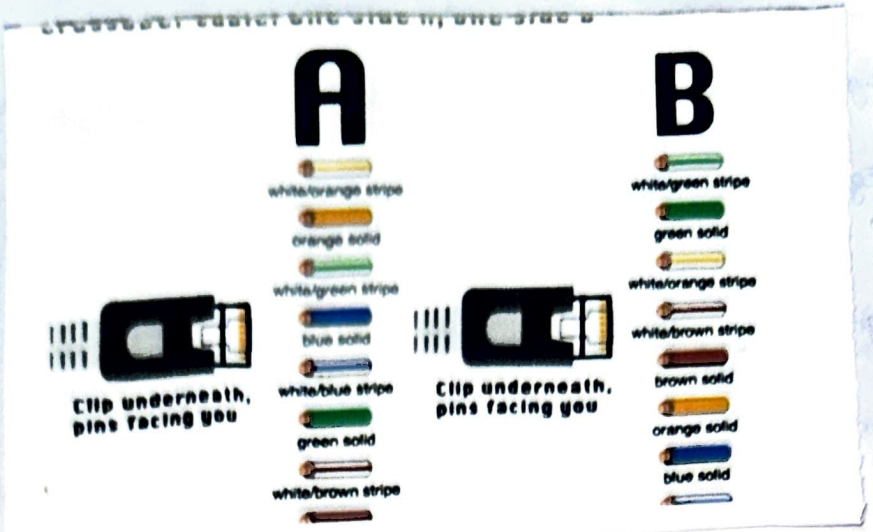
* Category 3 has 10bps data Transmission capacity
* Category 5 has upto 100 mbps data transmission
* category 5e has 1Gbps has capacity

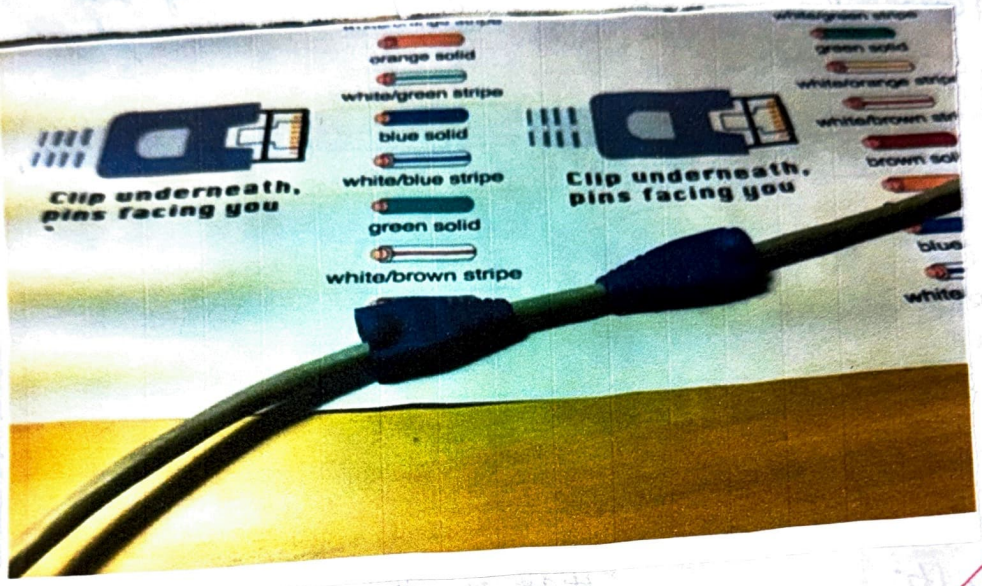

Foil Shield
Center Conductor
Outer Jacket
Dielectric

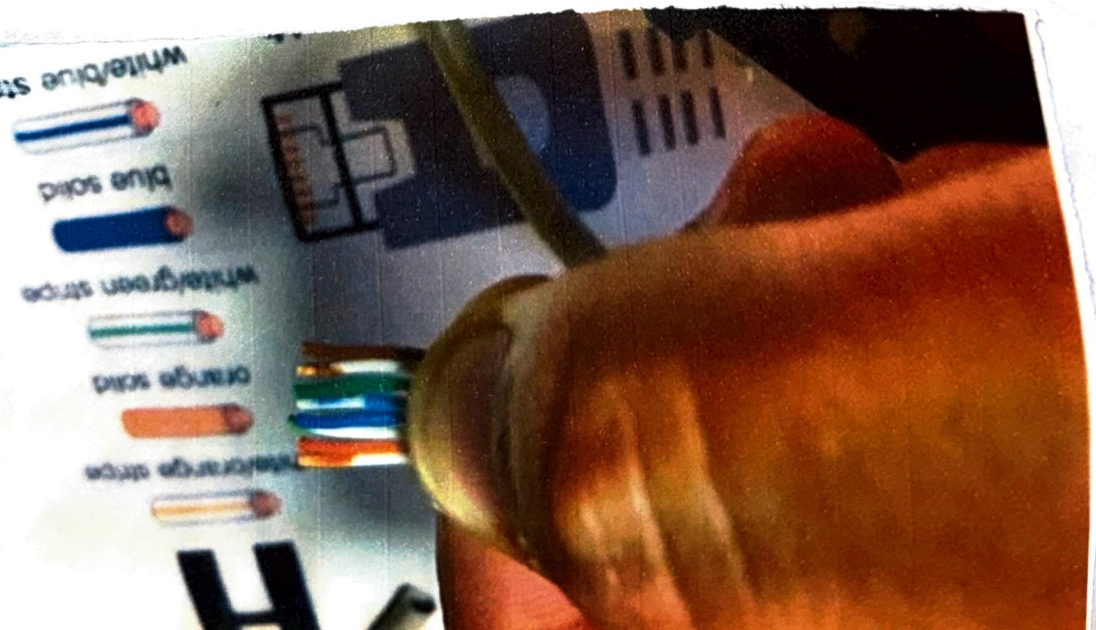



Straight thru cable
X-over cable
TX+ 1
TX- 2
RX+ 3
4
5
RX- 6
7
8
1 RX+
2 RX-
3 TX+
4
5
6 TX-
7
8
TX+1
TX -2
RX+3
4
5
RX- 6
7
8
1 TX+
2 TX-
3 RX+
4
5
6 RX-
7
8
Here you can compare the different pinsouts between a straight thru and x-over cable. The Transmit and Receive pins are also shown

Cable Type	Category	Maximum Data Transmission	Advantages / Disadvantages	Applications
UTP	Category 3	10bps	Advantages * Cheaper install * Easy to install	10 BASE-T ethernet
	Category 5	upto 100 mbps	They have small overall diameter	Fast ethernet, Gigabit
	category 5e	1 Gbps	More prone to EMI	
STP	Category 6, 6a	10Gbps	Advantages * Shielded * Faster * Less prone to noise	* Gigabit ethernet * Widely used in data centre
Coaxial Cable	RG-6 RG-59 RG-11	10-100 Mbps	Advantages * High bandwidth * Immune to interfe	* Speed of signal is 500 * High speed
Fibre optics cable	Single mode Multi mode	100Gbps	Advantages * High speed * High bandwidth * High security Disadvantage * Expensive	* Maximum distance of fibre optics cable is around 100m


A
white/orange stripe
orange solid
white/green stripe
blue solid
white/blue stripe
green solid
white/brown stripe
Clip underneath, pins facing you
B
white/green stripe
green solid
white/orange stripe
white/brown stripe
brown solid
orange solid
blue solid
Clip underneath, pins facing you


orange solid
white/green stripe
blue solid
white/blue stripe
green solid
white/brown stripe
Clip underneath, pins facing you
Clip underneath, pins facing you


blue solid
orange solid

Make your own Ethernet cross-over

Tools and parts needed:

* Either cabling cat 5e is certified for gigabit support but cat 5e cabling work as well
* A crimping tool
* Two RJ45 plugs
* Optional two plug shield

Step 1:

To start construction of the device begin by threading

Step 2:

Next strip approximately 1.5 cm of cable ~~thei~~ shielding from both ends.

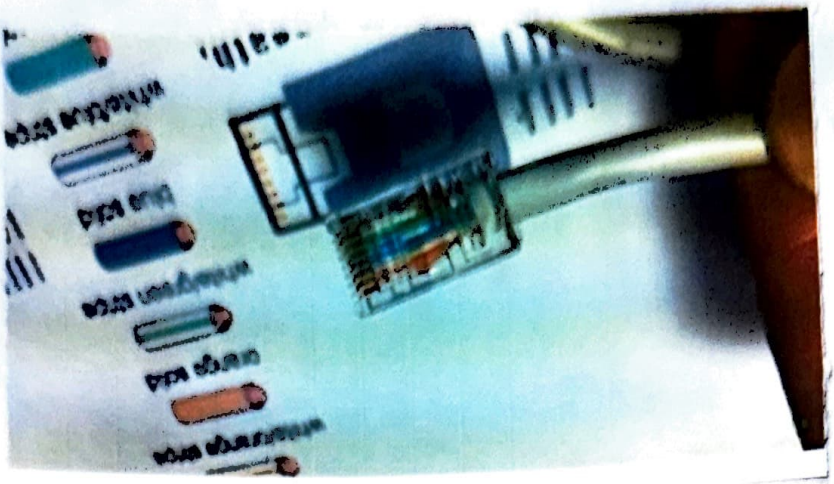
Step 3:

After you will need to untangle the wires, there should be four twisted pairs.

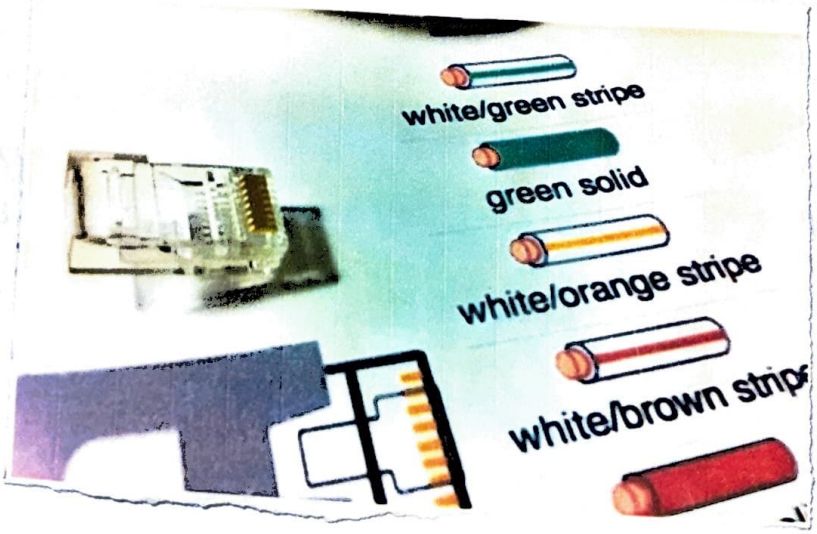
Step 4: Once the order is correct. Bunch them to gather in a ~~line~~ and if there are any that stick out farther than other strip them back to create an even level.

Step 5:

Next push the cable right in the not challenge to be just over the cable shielding and if it isn't that means that you stripped off too much shielding




A
B


white/green stripe
green solid
white/orange stripe
white/brown stripe

Step 6 : After wires are securely sitting inside the plug insert it into crimp - slot to go and push.

Step 7 : Lastly repeat for the other ends using diagram to test it plug it in attempt to connect two devices directly

Result : Thus, different types of network cables were studied successfully and straight through and cross over ethernet cable were tested successfully